

- LDAP Authentication
 - What is LDAP?
 - *Lightweight Directory Access Protocol*
 - Is like a digital "phonebook" for resources
 - <https://www.openldap.org/>
 - LDAP gives you the ability to query Directory Services
 - This is basically a network-based database of network resources
 - Files
 - Printers
 - Servers
 - Users
 - Is Windows Active Directory and LDAP the same thing?
 - No, but AD uses LDAP
 - How is it used for Authentication?
 - *ldapAuthProcess app*
 1. Connection Establishment:
 - The client, which could be a user or an application, initiates a connection to the LDAP server using the LDAP protocol (usually over port 389 or over SSL on port 636).
 2. Binding:
 - The client sends a "bind" request to the LDAP server. This request contains the user's distinguished name (DN) and a password. This can be an anonymous bind, a simple bind (username and password), or other supported authentication methods like SASL (Simple Authentication and Security Layer).
 3. Server Validation:
 - The LDAP server checks the provided credentials against the directory information. If the credentials are correct, the server responds with a success message. If not, it returns an error indicating authentication failure.
 4. Search (Optional):
 - Upon successful binding, the client may perform a search to retrieve additional information from the directory, such as user attributes or group memberships.
 5. Authentication Response:
 - If the server validates the credentials, the client receives confirmation and is granted access to the directory information or any services the user is authorized to access.
 6. Session Establishment:
 - After successful authentication, the user/client can establish a session to access the requested resources or services.